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f(x) 198

$$v(t) = \frac{dx}{dt} = f'(t)$$

$$\Rightarrow \int_{t_0}^t v(t) dt = f(t) - f(t_0)$$

$$\begin{aligned} \therefore f(t) &= f(t_0) + \int_{t_0}^t v(t) dt \\ &= x_0 + \int_{t_0}^t v(t) dt \end{aligned}$$

$$\Rightarrow f(t) = x_0 + \int_{t_0}^t v(t) dt .$$

total work $\leadsto \int_a^b v(t) dt$

TM $\leadsto \int_a^b |v(t)| dt .$